

Case Study 1: Design of a High-Speed Low-Power MOS Sequential Circuit

The objective of this design is to develop a 4-bit synchronous up/down counter for a low-power embedded processor using MOS-based sequential logic circuits. The counter must operate at high frequency, support both up and down counting using a control signal, include synchronous reset, and minimize both power consumption and delay. The overall architecture consists of four flip-flops connected in series, where each flip-flop represents one bit of the counter. A control signal is used to decide whether the counter increments or decrements. For high-speed and low-power operation, transmission gate flip-flops are preferred because they provide better signal integrity, reduced voltage drop, and lower power compared to pass transistor logic and pseudo-NMOS designs.

At the circuit level, each flip-flop is implemented using a transmission gate based master-slave configuration, which ensures proper edge-triggered operation. The next-state logic is designed using pass transistor logic or complementary pass transistor logic to reduce transistor count and improve speed. The state diagram represents all possible counting states, and the state table defines transitions based on the control input. The next-state equations are derived from this table and implemented using MOS logic circuits.

In timing analysis, important parameters such as setup time, hold time, and clock-to-output delay are considered. The critical path of the circuit typically lies in the combinational logic and flip-flop propagation delay. For power analysis, different flip-flop designs such as transmission gate based, static CMOS, and pseudo-NMOS are compared. Dynamic power is mainly due to switching activity, while static power arises from leakage currents and ratioed logic behavior in pseudo-NMOS circuits.

To optimize the design, techniques such as clock gating are used to reduce unnecessary switching activity, and careful transistor sizing is applied to maintain speed while minimizing power. Threshold voltage drop issues in pass transistor logic can be reduced by using transmission gates. Noise margins can be improved by using full CMOS structures where required. If the supply voltage is reduced, pass transistor logic suffers from threshold voltage loss and reduced signal swing, while transmission gate logic remains more reliable due to full voltage transfer capability. Overall, the design requires a balance between speed, power, and area to achieve optimal performance.

Case Study 2: Design of a High-Speed Low-Power SRAM with Sense Amplifier

The goal of this design is to implement a 64×8 SRAM memory block that supports fast read access, low power consumption, and reliable operation under process variations. The memory uses standard 6-transistor SRAM cells for data storage, along with supporting circuits such as row decoder, column selection logic, precharge circuit, and sense amplifier. The bit-lines are precharged to the supply voltage before each read operation to enable faster sensing.

The 6T SRAM cell consists of two cross-coupled inverters and two access transistors. During a read operation, the word line is activated, allowing the stored data to slightly discharge one of the bit-lines. In a write operation, data is forced into the cell by driving the bit-lines. During hold condition, the cell maintains its value using feedback. One important issue is read disturb, where the stored data may be affected during read; this is minimized by proper transistor sizing to ensure stability.

The sense amplifier is designed using a differential latch structure with cross-coupled inverters. It detects small voltage differences between bit-lines and amplifies them quickly using positive feedback. Even a small voltage difference is converted into a full logic level output due to the regenerative action of the latch. The timing of the circuit includes precharge phase, word line activation, and sense amplifier enable. The sense amplifier must be triggered only after sufficient voltage difference develops on the bit-lines.

At the circuit level, the precharge circuit uses PMOS transistors to charge bit-lines to supply voltage, and the sense amplifier is implemented using cross-coupled MOS transistors. Care is taken to avoid short-circuit currents and ensure proper isolation. Performance analysis includes read delay, which depends on bit-line discharge and sensing time, and power consumption, which includes precharge and switching power. Voltage-mode sensing and current-mode sensing are compared, where current-mode is faster but more complex.

Design challenges include process variations, transistor mismatch in the sense amplifier, and large bit-line capacitance. These can affect accuracy and speed. Improvements such as offset cancellation, reduced bit-line swing, and hierarchical bit-line structures help enhance performance. When supply voltage is reduced, the voltage difference on bit-lines decreases, making sensing more difficult. In such cases, current sense amplifiers are more robust compared to voltage latch type amplifiers. Overall, the design involves trade-offs between speed, power, and stability to achieve reliable SRAM operation.